Andrés Araque

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Education

University of Central Florida — Expected 2027

B.S. Mathematics

Coursework: Mathematical Foundations of AI & Machine Learning; Optimization Theory & Mathematical Modeling; Linear Algebra.

Selected Work

Ultralearn / Adaptive SAT math — In development

Building an adaptive tutor around a persistent learner graph: diagnose what a student knows, use that state to choose the next useful step, and space review across connected concepts. Living ink becomes the interface. @ultralearns.

Hacky / AI matchmaking — ~32 hours · Knight Hacks IX

Directed agentic development of the prototype, React interface, and conversational onboarding. Focused context, a custom knit companion, and carefully paced matching cards turn finding a teammate into a polished conversation. Built with a team of three.

Étude / Adaptive piano tutor — ~44 hours · Gator Hack V finalist

Owned the entire learning architecture: a 207-skill graph, diagnostic, Bayesian mastery updates, and spaced review. A constrained planner chooses practice from learner evidence and goals; code validates decisions and supplies a deterministic fallback. Built with a team of three.

Donna / Campus connections — On the App Store

Part of the founding development team, building UI and backend logic for a social app connecting people at UCF. Lead the current refactor toward an architecture agents can navigate and extend, with the right tools and context.

How I Build

I design agentic development systems that deliver ambitious, polished products at speed. The learning build for Étude used focused Claude Code worker sessions, durable context, separate PRs, browser verification, and CI.

- **Context and orchestration:** Claude Code, Codex, focused parallel workers, isolated Git worktrees, repository architecture, reusable skills, hooks, specs, and handoffs.
- **Quality and improvement:** Browser checks, tests, CI, and feedback loops that improve agent tools and instructions between runs. Authored Turn to Life and Skemify.

Product stack: TypeScript, React, Next.js, Vite, Electron, Python, PostgreSQL, Playwright, Vitest.

Work & Teaching

A3 Tutors Visit website — Founder & Tutor · 2024–present

I coach and tutor students for the SAT, helping them qualify for Bright Futures scholarships.

Hackathon times are elapsed team repository build windows, rounded. github.com/seeko-codes